

(KTXFI2EBNF) Physics II. Test 2 (Midterm Retake)
December 8, 2025

1. An electron has a kinetic energy of 0.5 keV.
 - (a) What is its velocity?
 - (b) What is its momentum?
 - (c) What is the wavelength of the associated de Broglie wave? (10 points)
2. A metal surface is illuminated with light of wavelength $9 \cdot 10^{-8}$ m. What is the velocity of the emitted electrons if the photoelectric effect starts at a wavelength of $2.88 \cdot 10^{-7}$ m?
($m_e = 9.1 \cdot 10^{-31}$ kg) (10 points)
3. In a Compton scattering experiment, X-rays with a wavelength of 0.2 nm are used.
($m_e = 9.1 \cdot 10^{-31}$ kg)
 - (a) At what scattering angle does the wavelength of the radiation increase by 1%?
 - (b) At what angle does the wavelength become 0.01% larger? (10 points)
4. A light source has a power output of 20 W.
 - (a) What is the frequency of the light if the entire energy of the source is emitted as light with wavelength $\lambda = 800$ nm?
 - (b) How many photons are emitted by the light source per second? (10 points)
5. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 60 ns; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is 5 ns. What is the propagation velocity of the particle beam? Determine the value of $\frac{m}{m_0}$ for the electron! Calculate the rest energy of an electron corresponding to its rest mass ($m_e 9.1 \cdot 10^{-31}$ kg) in joules and in electronvolts!
(1 MeV = $1.6 \cdot 10^{-13}$ J) (10 points)

Condition for obtaining a signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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